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Deep Learning and Ensemble Machine Learning-Based Approach to Detect Cyberbullying Using Twitter Data

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***Abstract*— The development of the modern web as well as its widespread use on communication sites like Facebook, YouTube, and Twitter have made life significantly easier when handling some very private issues. The goal of the paper is to reduce online abuse and establish a criticism-free virtual community. In this research, we have assessed a significant number of English tweets using several characteristics. We use natural language processing and machine learning (ML) approaches to convey clean information. Furthermore, the group has expanded the use of deep learning (DL) methodologies by applying tokenization with padding. Harassment that occurs over the internet can be swiftly identified through the use of natural language processing as well as mathematical visualization. A variety of classifiers have been utilized in our study. We increased the study by including several DL structures. It takes a lot of information to accurately identify acts of harassment. To quickly identify articles related to online abuse, we integrated a pair of data sets, resulting in 1,47,682 tweets about various viewpoints. We can notice which combination structure (MLP+SGD+LR) conducts better after comprehending the ML and DL approaches. Its multilabel assessments of accuracy, precision, and recall are all 99.11%.**

Keywords*— *Online Harassment, Stacking Model, Natural Language Processing, English Text-based Tweets, Machine Learning, Deep Learning.

I. INTRODUCTION

These days, online communication plays a major role in our daily activities. Social networking sites benefit us in numerous capacities, but they have additionally brought harm. [1]. We use online platforms for a range of purposes in everyday life, such as work, education, amusement, and private development [2]. Our generation's reliance on multimedia for daily tasks is growing as a result of modern technology as well as rapid internet access for significant development. Minorities can surf freely on the extensive internet. Age restrictions, insufficient monitoring techniques, and inadequate Sentiment Analysis safety protocols are common on social media websites. Nowadays, they have a regularly bad effect on young people and adolescents [3]. When the global web initially started, humans could utilize social networking sites using laptops, smartphones, and tablets, and the phenomenon was growing rapidly [4]. Since they frequently utilize social media sites and choose online activities over spending time with close companions, modern kids aren't being raised appropriately [5]. People can use an online connection system to exchange important information, images, and videos [6]. Online communication sites like Facebook, YouTube, Twitter, Instagram, TikTok, WeChat, Telegram, Snapchat, Reddit, and WhatsApp belong to the least popular ones globally [7]. Because of the variety of social networking sites that it utilizes and the privacy that ensues, online harassment can affect anyone, everywhere, at every moment. Online harassment is the act of using online platforms to further our negative objectives of intimidating, upsetting, or dehumanizing someone else. Nearly everyone, including kids, may encounter online harassment in everyday interactions from internet harassers. Sometimes, such individuals unknowingly startle people who don't realize how important the internet is and how they make use of it [8]. The following is an outline of the research paper's greatest purposes.

- Initially, we try to provide a summary of relevant research on identifying and categorizing detrimental tweets in online harassment.
- Additionally, we remove information according to several standards, such as eliminating unique symbols, links, numerous spaces, and punctuation.
- After the initial treatment of the information, we determined characteristics using a count vectorizer (CV) with term frequency-inverse document frequency (TF-IDF) for ML, and then for DL we used tokenization with padding.
- We execute every algorithmic classification with the associated mathematical formula after providing an in-depth description of all of them.
- A hybrid model based on ML operates better than the rest of the ML and DL algorithms since it evaluates efficiency measures to detect online harassment.

II. RELATED WORK

Numerous studies on the subject of determining online harassment have been conducted, with an emphasis on English language longevity in particular. The outcomes of these investigations provide strong support for the present

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study by demonstrating the requirement for effective strategies to stop online harassment on internet sites. The following subsection covers appropriate articles on multi-class Sentiment Analysis in English and other languages. We focused on key characteristics of Sentiment Analysis papers, such as the amount of information sets, the total amount of classes, the methods used, and the results. In reaction to several classes of Sentiment Analysis, various semantic viewpoints came to light recently.

Introducing a CNN and LSTM-based algorithm, Haque et al. [9] discussed the challenges of multi-class evaluation of thoughts in English online criticism, achieving 85.8% accuracy and 86% F1 scores around a set of 42,036 criticisms on Facebook. The proposed approach is superior to conventional approaches by proving its efficacy in the actual recognition of emotions using an internet-integrated application.

Using multiple languages of information, Chakravarthi et al. [10] proposed developing an individualized deep network model that could detect and encourage favorable reactions. The algorithm they used combined implementation compared to T5-Sentence with the CNN model, resulting in larger F1 scores of 75% for English, 62% for Tamil, and 67% for Malayalam.

When it comes to focusing on online harassment, Obaid et al.'s work [11] represents the outside as it draws attention to the risks that teenagers face but often ignore. The research they conducted uses an algorithm that uses DL that utilizes Twitter data and fuzzy reasoning to assess the severity of online harassment.

A technique for using ML techniques to recognize rude speech in Bangla was proposed by Mahmud et al. [12]. They achieved an accuracy percentage of 97% in identifying harassment in Bangla by applying transformed documents and logistic regression (LR).

According to Mahmud et al. [13], using ML-based models has greatly improved studies on online harassment using the Twitter dataset. LightGBM surpassed every other technique when outcome statistics were taken into consideration, yielding findings of 85% recall, 85.5% accuracy, 84% precision, and an 84.49% F1 score.

Akhter et al. [14] created a robust combination ML model with excellent precision rates of 98.57% and 98.82% in binary and multilabel verification to identify online harassment in English on the internet. The approach they use includes efficient content planning, data set adjustment using the particular case durability limits, and removing features with TF-IDF.

A previous study is shown with limitations in Table 1. We used six different ML algorithms and also used three different DL models, including certain combination methods that use stacking classification methods in addition to arranging to identify the most efficient outcome based on the most appropriate model, to cross any necessary limitations. Our approach analyzes massive amounts of data, helping to identify bullying assaults. It also comprises five categories, each of which has a different place.

TABLE I. RELATED PUBLICATIONS ALONG WITH BOUNDARIES.

References	Models	Dataset Size	Accuracy	Limitations
Haque et al. 2023 [9]	SVC, SGD, RF, LR, MNB, DT, CLSTM, BiGRU, BiLSTM, and LSTM	42,036	CLSTM: 85.80%	Detection Accuracy is lower.
Chakravarthi 2022 [10]	SVM, LR, KNN, DT, CNN	59,354	CNN: 75% (English), 67% (Malayalam), 62% (Tamil)	It is limited to specific languages and Sentiment Analysis. Detection Accuracy is also lower.
Obaid et al. 2023 [11]	LSTM	47,733	LSTM: 93.6%	Limited to specific language and dataset with limited accuracy.
Mahmud et al. 2023 [12]	LR, MB, DT, RF, SVM, AdaBoost, GB, SGD, ET, KNN, MLP	3000	LR: 97%	The dataset used for the study was relatively small the limited preprocessing approach was used.
Mahmud et al. 2022 [13]	LightGBM	47,000	LightGBM: 85.5%	Limited preprocessing approach used and detection accuracy is also lower.
Akhter et al. 2023 [14]	DT, RF, LR, and MLP	44,001	MLP: 98.82% LR: 98.57%	No DL models are used. Models' execution time is high.

III. MATERIALS AND METHODS

The remainder of this article describes our recommended methodology concerning the various DL and ML algorithms built into the architecture. Let's start by going over how the idea works. Following this, a quick rundown of the ML and DL algorithms is provided. The method of operation of our suggested approach is shown in "Fig. 1". First, we collected data through two publicly available databases of suggestions made in English on online platforms. After the fact that we brought them together through just one dataset as well as divided it into six categories: Religion, Age, Gender, Ethnicity, Other cyberbullying, and Not cyberbullying. To make it easier to prepare the information, the content was subsequently filtered to eliminate English stop phrases, internet links, symbols, unique indications, and numbers. We utilize tokenization with padding to supply our DL algorithms and a Count Vectorizer along with a TF-IDF converter for creating our ML classifiers from textual information. To check for possible disparities in classes throughout both testing and training, we recalculate the information using the Instance Hardness Threshold (IHT). We divided the data using k-fold cross-validation for testing and training. ML classifiers like Staking, SGD, RF, MLP, Bagging, Adaboost, and K-NN. DL algorithms like BiGRU, CNN, and CLSTM have been used as well for the comparison study. And evaluated achievement utilizing conventional assessment indicators like F1 score, recall, accuracy, and precision. Ultimately, we used the top-performing algorithms as the foundation for an internet-based application.

A. Dataset Collection

Gathering of information, the collection of data utilized in this study for detecting cyberbullying was gathered using the Kaggle website. Over 47,000 of the tweets that were taken during COVID-19 have been incorporated into the first publicly accessible dataset that we have. 87% of people reported having witnessed online harassment during that

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time, and 36.5% reported having experienced it themselves, according to UNICEF statistics [15]. About 100,000 tweets were collected from social media platforms for the second dataset. A multi-class classification method is used to label it.

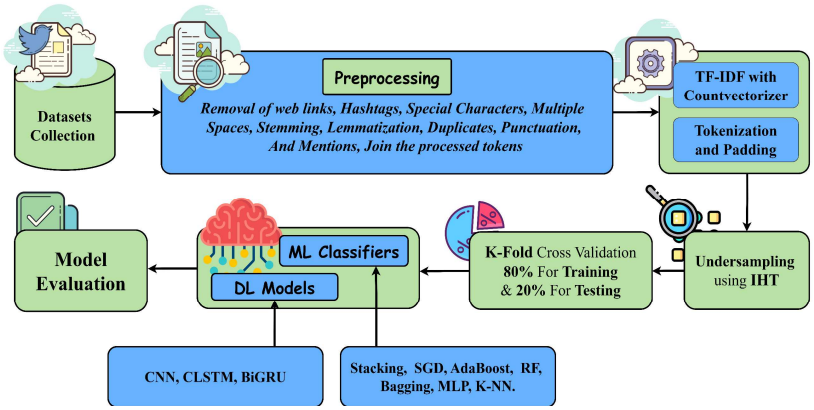


Fig. 1: Workflow of our Proposed Model.

50,000 instances of non-cyberbullying 17,000 instances of ethnicity-based cyberbullying 17,000 cases of gender-based cyberbullying and approximately 16,000 cases of religious cyberbullying [16]

We combine the two sets of data, which are commonly used by scholars; however, a review of prior research on this subject reveals an excessive number of constraints. Some writers don't collect as much data; therefore, they can't identify every bullying text or comment. We will require more knowledge that is, vast amounts of information to get past those limitations. Data collection will provide us with that knowledge. We incorporate data into our job for this reason. The data was then split, with 20% going toward testing and the remaining 80% going toward training. The dataset's six categories along with an estimate of how many tweets belong in each category are visualized in this pie chart with the proportion of the tweets for all of the categories shown in “Fig. 2”.

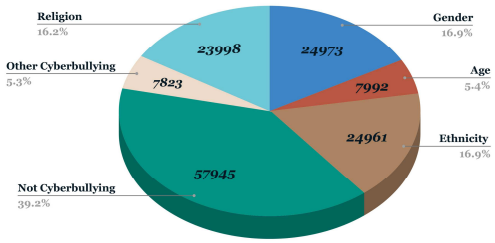


Fig. 2. A pie chart shows the percentage of tweets for each category.

B. Data Preprocessing And Feature Extraction

Getting ready for raw information and overestimation an algorithm has grown into the new standard for information initial processing. That is among many crucial procedures during the framework-development procedure. In the paper's initial evaluation, our team handled deficient characters replacing an approach that ensures they are classified according to features, spaces, mentions and links, punctuation, stop phrases, and the usage of new accordance characters are all removed. Additionally, we eliminate contractions and utf8 and ASCII symbols such as '\x9a \x91 \x97 \x9a \x97'. Just the "#" symbol should be removed from the end of the text, the middle hashtags should remain in place. Remove special characters from your text that don't belong there, such as "&" and "$.". After that, all typical and ending words in English are eliminated.

For feature extraction on ML, we set CV and TF-IDF [17] as combined to produce a potent written analysis method based on equation 1. The TF-IDF converter afterward applies its weighting to this structure, taking into account the quantity and uniqueness of every value throughout the information. By providing an improved comprehension of linguistic significance in materials, this combination of methods enhances ML classifiers for obligations like online harassment evaluation and categorization, and also applications involving natural language processing. Tokenizing the document, padding at each word stage, and symbol allocation for each word were done to ensure that the DL models' categorization algorithms functioned.

$$R_M = Tf\,(Cv\,(value)) \qquad 1$$

C. Machine Learning Classifiers

The study determines outcomes employing 6 classification algorithms for ML (Staking, SGD, RF, MLP, Bagging, Adaboost, and K-NN) to develop a system to identify English assault. The effectiveness of every algorithm has been assessed using a range of efficiency metrics. We are going to look over the various machine-learning approaches that were employed in the present research for predicting and classifying the models that are described in this part of the paper.

Group instruction is a combined machine-learning technique that takes into account the predictive skills of multiple fundamental models to increase predictability [18]. Each beginning strategy can utilize methods based on ML. If an assembly is formed with a consistent mix of build respondents, which is frequently called a standard structure for guidance, in addition, the resultant combined is called numerous. Integration growth uses a variety of techniques: stacking, boosting, and bagging. To improve the model's productivity, stacking has evolved into a team process that makes use of a meta-learner system in addition to utilizing methods for categorization based on the fact that they have

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been created on an equivalent database. In this study, multiple ML techniques are first implemented in a one-level organizing mechanism. Finally, by combining LogitBoost with the predictions generated by every classification system, completed projections can be obtained. The configuration of the stacking group is displayed in “Fig. 3”.

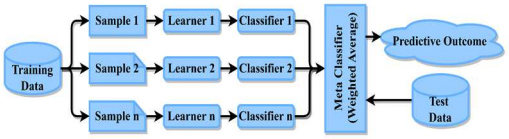


Fig. 3: Stacking model working flow.

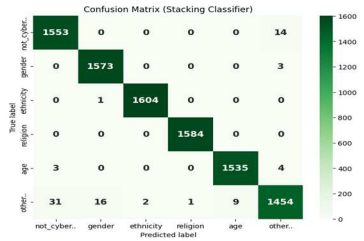
D. Deep Learning Models

This study additionally utilizes three different DL models for evaluating efficacy: BiGRU, CNN, and CLSTM. The goal is to develop a structure to identify English bullying assaults. The efficiency of every system has been assessed using a range of outcome signs. We are going to go over the various DL approaches that were applied to the classification approaches and estimation structure for the present research.

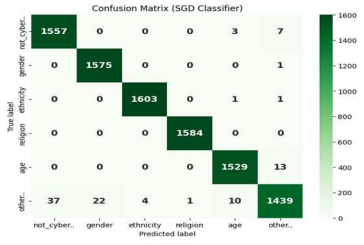
A popular variant that utilizes commonly employed direct neural networks is CNN. It was created to solve 2D issues. The previously mentioned CNN is comparable to the simulations that proceed, but instead of the entirely related wide level, a component made up of long short-term memory (LSTM) has been added. The GRU combined candidate private indicates and alters an LSTM's three entries by recovering and changing entries to control the flow of content at different stages [19].

IV. Experimental Results And Discussions

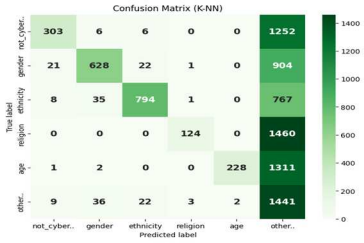
The manufacturing process of several classes and the categorization of the similarly suggested DL and ML techniques discussed above were applied to correspond with the method of learning various tweets that included information to determine and categorize various forms of internet abuse classes using the assessed information. We utilized a range of managed methods for categorizing, and measurement metrics like accuracy, precision, recall, and F1 score were utilized to evaluate the theories. The confusion matrices employed to compute these numbers are shown in “Fig. 4”. Six ML methods with a hybrid model have been developed and equipped with the provided dataset to facilitate categorization. As seen in “Fig. 4(a)”, the hybrid stacking model effectively categorizes 9,303 tweets, while its structure erroneously categorizes simply 84 tweets. As shown in “Fig. 4(a)”, the hybrid algorithm achieved a 99.11% accuracy in classifying. Information error is uncommon and Stacking (SGD+RF+MLP), MLP, SGD, RF, K-NN, AdaBoost, and Bagging categorization are all usually precise. “Fig. 4(b)”, refers to SGD classifiers that stand with a good classification result as an individual classifier which is 98.93%. “Fig. 4(c)” k-NN confusion matrix, where k-NN shows a lowest accuracy score of 37.48%. We also applied three methods based on DL that have been prepared for purposes of categorization to the provided information set. As shown in “Fig. 4(d)”, the CNN simulation achieved a 91.64% classification accuracy. “Fig. 4(e)” makes it clear that data incorrect categorization is mostly uncommon and that BiGRU categorization are typically reliable.



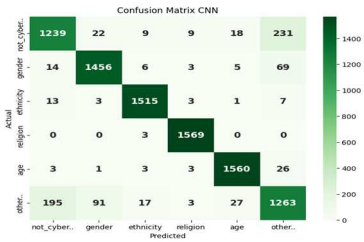
a. Stacking Classifier



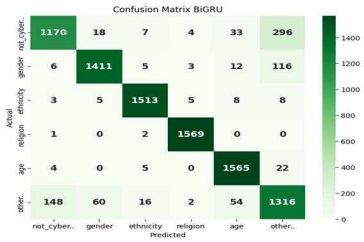
b. SGD



c. KNN



d. CNN



e. BiGRU

Fig. 4. ML and DL algorithms matrix of confusion.

The total 10 models with hybrid model outcomes are listed in Table 2 and include the F1 score, recall, accuracy, and precision score. Similar information is shown utilizing a bar chart in “Fig. 5” for ML classifiers and “Fig. 6” for DL models. The precision, recall, and F1 scores of ML as well as DL algorithms were the primary indicators of effectiveness

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197

used to evaluate them. Great precision was shown by SGD+RF+MLP(Stacking) and SGD, which scored 99.11% and 98.94%, respectively. The recall was also impressive, coming in at 99.11% and 98.93%, and both had F1 scores of 99.10% and 98.93%. The accuracy of both of these simulations was very high, at 99.11% and 98.93%, respectively. Powerful precision of 98.64% and 98.06% was also demonstrated by MLP and Random Forest Classifier, recall of 98.64% and 98.06% for both, and F1 score of 98.64% and 98.05%, respectively, achieving accuracies of 98.64% and 98.06%. The KNN classifier spotted an accuracy of 37.48% due to its lower recall and F1 score. In contrast, the K-NN, AdaBoost, and Bagging classifiers achieve lower accuracy scores of 37.48%, 71.17%, and 67.71% respectively. On the DL models, we acknowledge CNN, CLSTM, and BiGRU algorithms demonstrated harmonious outcomes, with accuracies exceeding 91% approximately. A more thorough evaluation of the algorithm's effectiveness is provided by the precision, recall, and F1 score; SGD+RF+MLP(Stacking) and SGD were the algorithms that performed best in each of these classifications. Precision indicates favorable accuracy in predictions and is measured as the ratio of actual positive predictions to the overall calculated favorable aspects. Recall also referred to as responsiveness or true positive rate, measures the modeling system's capacity to record every authentic certain illustration. The F1 score offers a fair assessment of an algorithm's effectiveness and is calculated as the harmonious average of precision and recall. In essence, recall highlights the system's capacity to detect all true positives, precision indicates how accurate the system is at estimating beneficial cases, and the F1 score is a combination of the two, providing a comprehensive evaluation of an algorithm's effectiveness in process categorization.

TABLE II. ML AND DL MODELS CLASSIFICATION RESULTS.

	Model	Accuracy	Precision	Recall	F1 Score
Machine Learning Models	SGD+RF+MLP (Stacking)	99.11	99.11	99.11	99.1
	MLP	98.64	98.64	98.64	98.64
	SGD	98.93	98.94	98.93	98.93
	R. Forest	98.06	98.06	98.06	98.05
	K-NN	37.48	81.55	37.48	37.66
	AdaBoost	71.17	76.46	71.17	70.23
	Bagging	67.71	84.83	67.71	65.77
Deep Learning Models	CNN	91.64	91.56	91.64	91.59
	CLSTM	91.29	91.31	91.29	91.26
	BiGRU	91.02	91.22	91.02	91.01

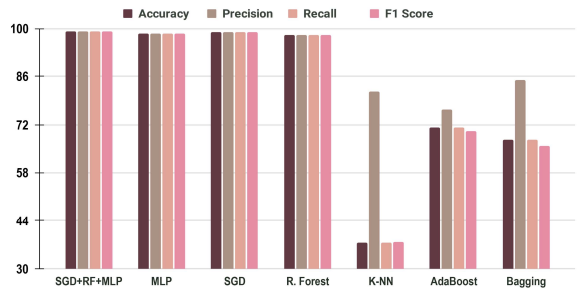


Fig. 5. Visualization of ML classifiers classification results.

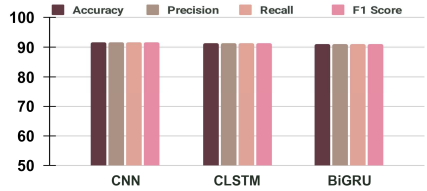


Fig. 6. Visualization of DL models classification results.

The efficiency of multiclass categorization is represented graphically and calculated using Receiver Operating Characteristic (ROC) curves. Depicting the balance between awareness and specificity at various thresholds helps visualize how well an approach can distinguish between distinct groups. Additionally, we attempt to illustrate the ROC curve for a several-class information set using various classifiers in “Fig. 7” below. Plotting TPR (Sensitivity) toward FPR (1-Specificity) for various threshold values is represented by the ROC curve. Generally, it starts at (0,0), which denotes a threshold where all things are categorized as negative, and ends at (1,1), which denotes a threshold where all things are categorized as positive. The trade-off between correctly classifying negative circumstances as positive and incorrectly classifying positive cases as positive is illustrated by the curve. Here “Fig. 7(a)” refers stacking model multiclass categorization ROC curve as well and “Fig. 7(b)” shows us the CNN model categorization of the ROC curve.

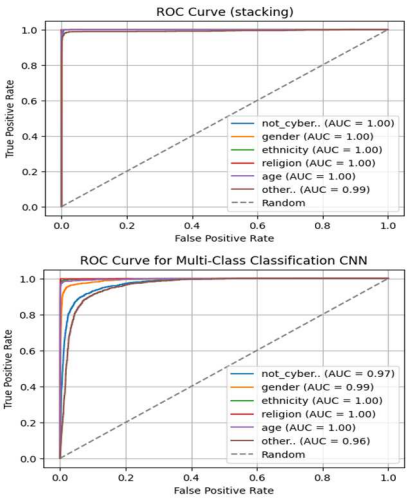


Fig. 7. ROC curve of best ML and DL approach.

Next, we assessed each ML or DL model's accuracy, precision, recall, and F1 score matrix to find out which one is most effective at identifying online bullying. “Fig. 4” and Table 2 present the results of the models' performance as determined by the F1 score, recall, accuracy, and precision. Accuracy alone is insufficient to evaluate a model's efficacy. As a result, we must select the best model from the seven ML and three DL models that are available for identifying online harassment. Recall and F1 score are also crucial considerations in this recession. Precession and recall are required in an equation that can be used to determine the final F1 score. Upon analyzing each matrix output, we discovered that each classifier performed essentially in the same way, with the stacking model outperforming the others. At that

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198

point, stacking outperformed the other competitors in terms of accuracy and recall achieving a 99.11% accuracy rate.

Concerning the precision SGD+RF+MLP (Stacking) matrix, the increased effectiveness is 99.11%. Subsequently, SGD+RF+MLP(Stacking) outperformed all the remaining algorithms, and the KNN classifier executed poorly when seven machine learning (ML) classifiers were analyzed using the Twitter data set with the aim of identifying online harassment. Comparing the efficacy of the proposed approach with findings from past studies is a vital phase in studies, especially in the fields of machine learning (ML), statistical analysis, and additional data-driven disciplines, as Table 3 illustrates. This evaluation aids in our understanding of our approach's efficacy relative to other, already-released versions as well as that of researchers. Regarding Table 3, the recommended approach has the greatest accuracy in forecasting for several classes of online harassment verification when contrasted with earlier studies. The proposed research used the SGD+RF+MLP (Stacking) algorithm and achieved a 99.11% median accuracy for classification.

TABLE III. COMPARING PREVIOUS WORK WITH OURS.

References	Feature Technique	Dataset	Classes	Accuracy	Precession	Recall	F1 Score
[9]	TF-IDF, CV & WE	42,036	4	85.80	86.00	85.00	86.00
[10]	TF-IDF, CV & WE	59,354	2	--	71.00	81.00	75.00
				--	64.00	65.00	62.00
				--	81.00	75.00	67.00
[11]	WE	47,733	2	93.67	--	93.62	93.64
[12]	TF-IDF	3,000	2	97.00	95.00	95.00	95.00
[13]	TF-IDF	47,000	6	85.50	84.00	85.00	84.49
[14]	TF-IDF	44,001	5	98.82	98.86	98.89	98.88
			2	98.57	98.58	98.56	98.56
Our Work	TF-IDF, CV & Tokenization	1,47,682	6	99.11	99.11	99.11	99.10

V. CONCLUSION AND FUTURE WORK

The analysis focuses on multi-class harassing behavior related to English text-based tweets on the internet, a problem with limited studies. Concentrating on the challenges with verification brought about by offenders hiding from disagreements. Content preparation, integration of methods for extracting features for ML classification algorithms as well as symbolization across DL algorithms, and information control utilizing IHT are all recommended components of an integrated ML approach to identifying online harassment. DL algorithms (CNN, CLSTM, and BiGRU) and ML methods SGD+RF+MLP(Stacking), MLP, SGD, RF, K-NN, AdaBoost, and Bagging are used together with a k-fold cross-validation method, and their effectiveness is assessed utilizing various metrics. Utilizing information from two publicly available information sets, we classified 1,47,682 online tweets into six categories. Through the study, we showed that by effectively analyzing and obtaining characteristics utilizing various techniques, multi-class online harassment has been capable of achieving a remarkable amount of accuracy across the syntax. With an accuracy rate of approximately 99.11%, using a mixed machine learning approach, the algorithm successfully identified online assaults on various social networking systems. It also achieved 91.64% validity using a deep learning method called CNN. We want to look into transformer-based methods. Techniques such as BERT, RoBERTa, and ELECTRA, among others, may find use in multiple languages in internet harassment files that encompass a range of online harassment categories.

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